INCH-POUND

MIL-P-63196A (AR)
AMENDMENT 3
16 March 1995
SUPERSEDING
AMENDMENT 2
26 August 1992

MILITARY SPECIFICATION

POLYURETHANE ELASTOMER BINDER
(For Use in Ammunition)

This Amendment forms a part of Military Specification MIL-P-63196A (AR) dated 27 May 1980 and is approved for use by the U.S. Army Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1 Change "two" to "three". Add "Type III: Polyurethane with Talc".

* 2.1 Delete: "MIL-H-48358 (AR) Amendment 2, 8 April 1988 HMX/Resin Explosive Composition LX-14-0 (For use in ammunition)".

PAGE 2

3.1 Delete in its entirety and substitute:

"3.1 Physical form. The polyurethane elastomer shall be supplied in granular, flake, or pelletized form. Type I shall be dusted with approximately 0.1% of calcium stearate (JAN-C-362), Type II with approximately 0.1% of commercial talc (MIL-T-50036), and Type III with approximately 0.5% of commercial talc (MIL-T-50036) to facilitate handling. Type III may also contain up to 0.02% Treadlube which is a potassium soap of coconut oil. The certificate of compliance furnished with the shipment shall certify which dusting agent has been used. For Type III orders the concentration of the Treadlube residue shall be provided."

PAGE 3

3.1.2 Viscosity: Delete in its entirety and substitute the following:

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"3.1.2 Viscosity. Solution viscosity of the polyurethane elastomer shall be between 180 and 900 centipoises when tested as described in 4.5.2."

3.1.2 Change "180" to "225".

3.1.3 Physical Properties: Delete in its entirety.

3.1.7 Add new paragraph:

"3.1.7 Age. The polyurethane elastomer shall not be older than 90 days from the date of manufacture when shipped."

PAGE 5

4.5.1.1 Delete in its entirety and substitute:

"4.5.1.1 Preparation of test solution. Weigh accurately a 45 ± 0.09 gram portion of the sample and transfer this sub-sample into a sealable pint jar (Note: If necessary, cut the sample into pieces approximately 1 cm long). Add 255 ± 0.05 grams of methyl ethyl ketone (MEK) to the jar and seal. Dissolve the contents of the jar on a roller mixer for a period not to exceed 24 hours. Care should be taken to prevent the loss of the solvent during this procedure. Use this solution for the solubility (4.5.1.2) and viscosity (4.5.2) determinations."

PAGE 6

4.5.2 Change "25 degrees C" to "23 degrees C" and add the following sentence after "... a No. 2 spindle":

"Prior to measuring the sample's solution viscosity, the calibration of the viscometer shall be verified using 200 and 500 centipoise standards. The viscosity standards shall be prepared in accordance with ASTM D 2162".

4.5.3 Correct "o" to "of", "overn" to "oven", and "Hq" to "Hg".

4.5.4 Correct "the" to "The".

PAGE 8

6.5 Change the first sentence from "Materials...B.F. Goodrich Chemical Co." to "Materials that have been found satisfactory for use in explosive compositions as a binder: Type I: Estane 5702F1, Type II: Estane 5703F1 and Type III: Estane 5703P manufactured by B.F. Goodrich Chemical Co.

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